Yuchen Wang

Ph.D. Student in Computer Science

Department of Computer Science, William & Mary

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Education

William & Mary Sep. 2024–Present
Ph.D. in Computer Science Department of Computer Science

Shanghai Jiao Tong University Sep. 2021–Mar. 2024
M.S. in Control Science & Engineering School of Electronic Information and Electrical Engineering

North China Electric Power University Sep. 2017–Jul. 2021
B.E. in Automation School of Control and Computer Engineering

Research Areas

World action models; world model-based reinforcement learning; physics-informed machine learning.

Selected Publications

[1] **Yuchen Wang***, Jiangtao Kong*, Sizhe Wei, Xiaochang Li, Haohong Lin, Hongjue Zhao, Tianyi Zhou, Lu Gan, and Huajie Shao. WestWorld: A Knowledge-Encoded Scalable Trajectory World Model for Diverse Robotic Systems. *International Conference on Machine Learning (ICML)*, 2026. **Spotlight** (2.2%).

[2] **Yuchen Wang**, Hongjue Zhao, Haohong Lin, Enze Xu, Lifang He, and Huajie Shao. A Generalizable Physics-Enhanced State Space Model for Long-Term Dynamics Forecasting in Complex Environments. *International Conference on Machine Learning (ICML)*, 2025.

[3] Zhenyu Zong, **Yuchen Wang**, Haohong Lin, Lu Gan, and Huajie Shao. A Generalizable Physics-guided Causal Model for Trajectory Prediction in Autonomous Driving. *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

[4] Hongjue Zhao, **Yuchen Wang**, Hairong Qi, Zijie Huang, Han Zhao, Lui Sha, and Huajie Shao. Accelerating Neural ODEs: A Variational Formulation-Based Approach. *International Conference on Learning Representations (ICLR)*, 2025.

All Publications

[1] Zhenyu Zong, **Yuchen Wang**, Jiangtao Kong, Xiaochang Li, Lui Sha, and Huajie Shao. An Adaptive Slow-Fast Framework for Real-Time Anomaly Detection in Autonomous Driving. *Conference on Robot Learning (CoRL)*, 2026.

[2] Enze Xu, Toon Tran, Hongjue Zhao, **Yuchen Wang**, Mengdi Huai, and Huajie Shao. Identifying Invariant Physical Dynamics Across Multiple Environments. *Transactions on Machine Learning Research (TMLR)*, 2026.

[3] Xiaochang Li, Chen Qian, Qineng Wang, Jiangtao Kong, **Yuchen Wang**, Ziyu Yao, Bo Ji, Long Cheng, Gang Zhou, and Huajie Shao. Lens: A Knowledge-Guided Foundation Model for Network Traffic. *Transactions on Machine Learning Research (TMLR)*, 2026.

[4] Hongjue Zhao, **Yuchen Wang**, Hairong Qi, Jiajia Li, Lui Sha, Han Zhao, and Huajie Shao. Accelerating Neural Differential Equations for Irregularly-Sampled Dynamical Systems Using Variational Formulation. *ICLR Workshop on AI4DifferentialEquations in Science*, 2024.

[5] **Yuchen Wang**, Can Xiong, Changjiang Ju, Genke Yang, Yu-wang Chen, and Xiaotian Yu. A Deep Transfer Operator Learning Method for Temperature Field Reconstruction in a Lithium-Ion Battery Pack. *IEEE Transactions on Industrial Informatics*, 2024. (IF=**11.7**).

[6] **Yuchen Wang**, Can Xiong, Yiming Wang, Po Xu, Changjiang Ju, Jianghao Shi, Genke Yang, and Jian Chu. Temperature State Prediction for Lithium-Ion Batteries Based on Improved Physics-Informed Neural Networks. *Journal of Energy Storage*, 2023. (IF=**8.9**).

Preprints and Submissions

[1] Hongjue Zhao, Yizhuo Chen, **Yuchen Wang**, Hairong Qi, Lui Sha, Tarek Abdelzaher, and Huajie Shao. Understanding the Theoretical Foundations of Deep Neural Networks through Differential Equations. *arXiv preprint*,

2026.

Research Experience

Research Assistant Sep. 2024–Present
William & Mary Advisor: Prof. Huajie Shao

- Co-developed a **Mixture-of-Experts (MoE) world model** with system-aware routing and morphology-aware embeddings, scaling pretraining to **89 robotic environments** and improving zero-/few-shot trajectory prediction and model-based control.
- Developed **Phy-SSM**, integrating partially known physics into **state-space models** to improve long-term dynamics forecasting from noisy, irregularly sampled data across three real-world applications.
- Contributed to a **physics-guided causal model** for **motion prediction in autonomous driving**, combining scene disentanglement with a kinematics-informed Neural ODE decoder to improve zero-shot generalization to unseen cities.
- Contributed to **VF-NODE**, a variational formulation-based method that bypasses numerical ODE solvers during training, achieving **10–1000× training speedups** over NODE-based baselines with comparable or higher accuracy.

Research Assistant Sep. 2021–Jan. 2024
Shanghai Jiao Tong University Advisors: Prof. Changjiang Ju and Prof. Genke Yang

- Developed a **physics-informed LSTM** with multi-head attention and learnable physical parameters for battery **thermal dynamics prediction**, improving generalization across operating conditions and battery aging.
- Developed a **deep transfer operator learning** framework combining Transformers, thermal PDE constraints, and **domain-adversarial learning** to reconstruct battery temperature fields under unseen operating conditions with limited training data.
- Designed a production-planning sub-algorithm for the National Key Research and Development Program of China project on factory operating systems for intelligent manufacturing in process industries.
- Developed algorithms for lithium-ion battery state of health, temperature, and safety estimation in a Ningbo City-funded battery systems project.
- Completed graduate thesis: *Research on Key Technologies for Temperature Field Reconstruction in a Lithium-Ion Battery Pack Based on Physics-Informed Machine Learning.*

Research Assistant Mar. 2020–Mar. 2021
North China Electric Power University Advisor: Prof. Tongxiang He

- Applied **PID control** in Single-Input-Single-Output and Cascade Three-Input-Three-Output setups for boiler water-level and feedwater-flow management.
- Developed Distributed Control System configuration diagrams using PAS-300 software.
- Completed undergraduate thesis: *A Study on Superheated Steam Temperature System Using the State Feedback Control with Error Integral Based on Simulink.*

Work Experience

Algorithm Engineer, Battery Management System Jun. 2022–Jul. 2023
Research and development role Advisor: Dr. Can Xiong

- Led development of an LSTM-based algorithm for lithium-ion battery temperature-state estimation.
- Contributed to an Extended Kalman Filter-based algorithm for lithium-ion battery state-of-charge estimation.
- Participated in a Gaussian Process Regression-based algorithm for lithium-ion battery state-of-health prediction.

Intern Jun. 2021–Sep. 2021
Ningbo Industrial Internet Institute Advisor: Prof. Changjiang Ju

- Developed industrial PLC HMI software in C#, supporting firmware updates and file operations.
- Participated in the preparation of a research report on the hydrogen energy industry.

Honors and Awards

- ICML Gold Reviewer Award (top 25%), 2026
- Outstanding Graduate of Shanghai Jiao Tong University, 2024
- Graduate Student Merit Scholarship, 2023
- Shanghai Jiao Tong University Second Academic Scholarship, 2022
- College-Level Third-Class Scholarship, North China Electric Power University, 2020
- Alumni Scholarship, 2019